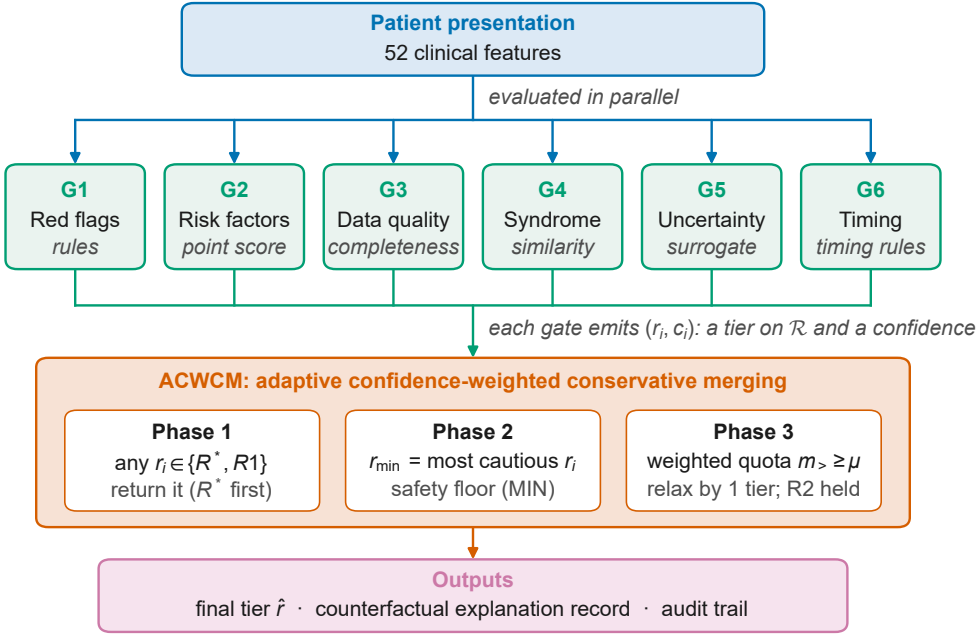

Patient presentation
52 clinical features
evaluated in parallel
G1
Red flags
rules
G2
Risk factors
point score
G3
Data quality
completeness
G4
Syndrome
similarity
G5
Uncertainty
surrogate
G6
Timing
timing rules
each gate emits (r_i, c_i): a tier on $\mathcal{R}$ and a confidence
ACWCM: adaptive confidence-weighted conservative merging
Phase 1
any $r_i \in \{R^*, R1\}$
return it (R^* first)
Phase 2
$r_{\min}$ = most cautious r_i
safety floor (MIN)
Phase 3
weighted quota $m_> \geq \mu$
relax by 1 tier; R2 held
Outputs
final tier $\hat{r}$ · counterfactual explanation record · audit trail